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# INTRODUCTION TO MEDICAL IMAGING

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SE374, Fall 2026

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University of Information and Technology,  
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# EVALUATION

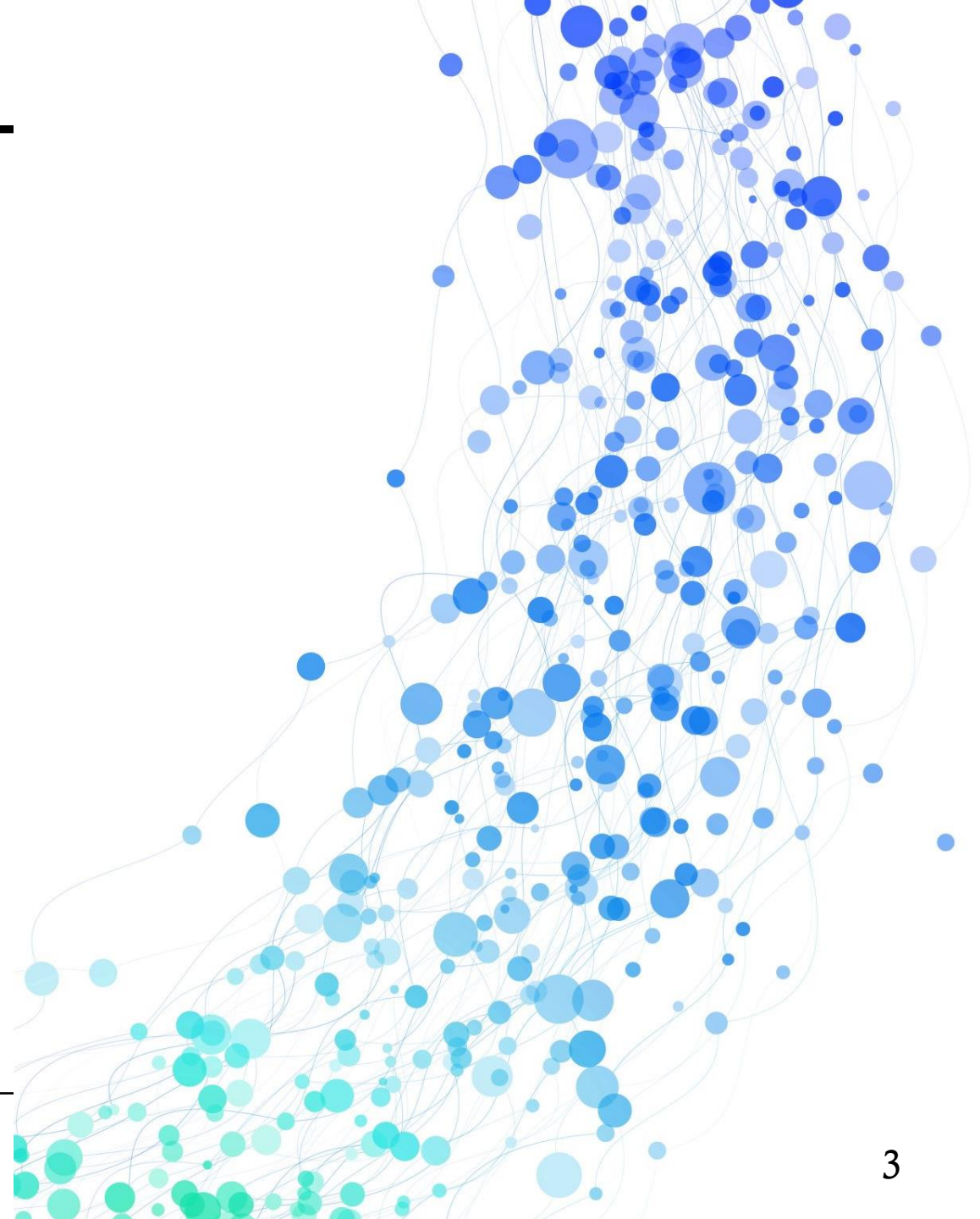
- Quiz (Attendance): 10%
- Lab: 30%
- Seminar + Project: 60%



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# LECTURE PLAN

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# LECTURES

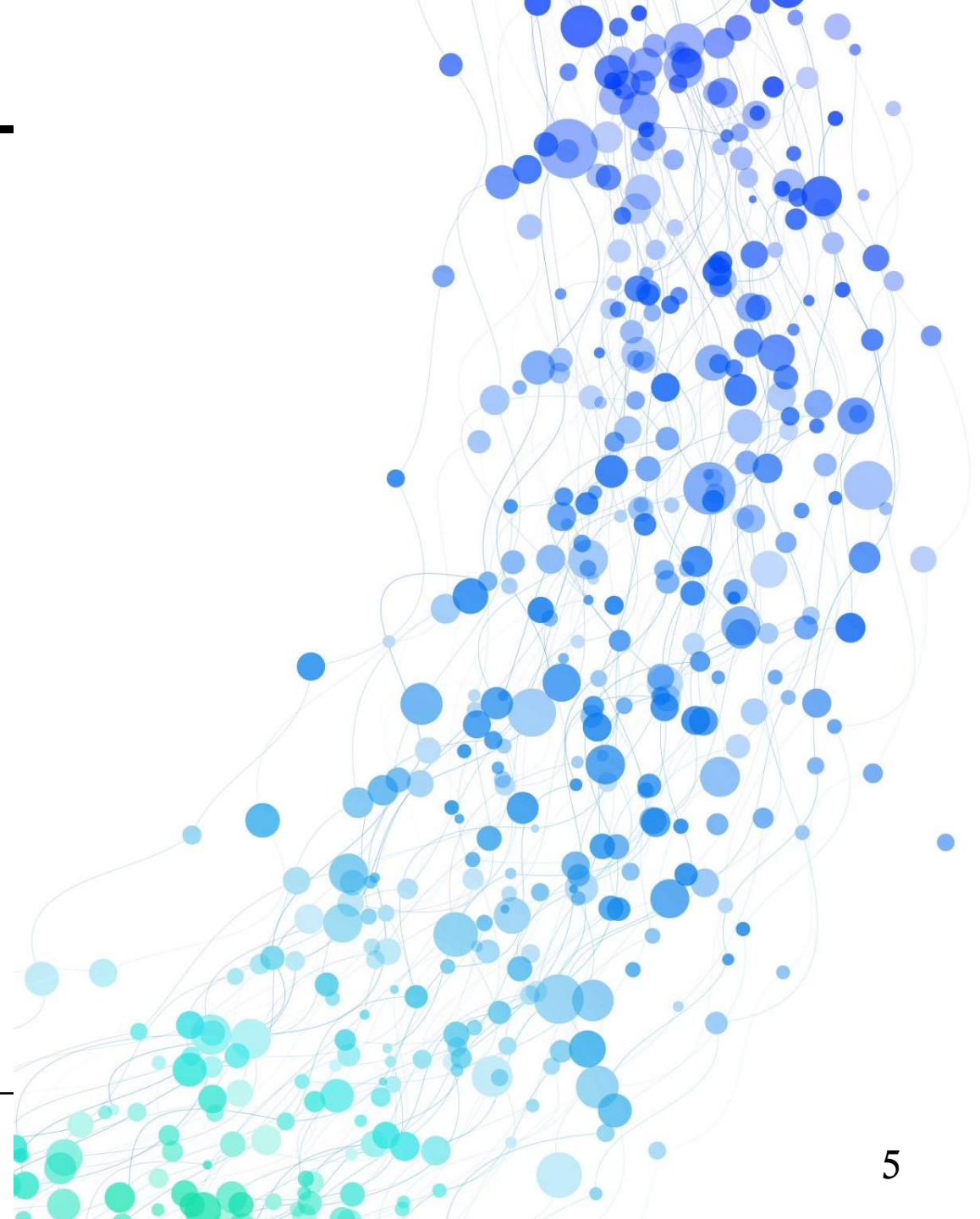
- Sessions 1-2: Introduction to common types of medical images.
- Session 3: Major diagnostic tasks and evaluation metrics.
- Sessions 4-5: Histopathological image processing.
- Session 6: Applications for predictive tasks using histopathological images.
- Sessions 7-8: CT/MRI image processing.
- Session 9: Applications for predictive tasks using CT images and CT reconstruction from sparse-view sinogram data.
- Sessions 11-12: Design and testing of diagnostic software systems.
- Sessions 13-15: Seminars on modern topics in medical image processing.



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# COMMON TYPES OF MEDICAL IMAGES

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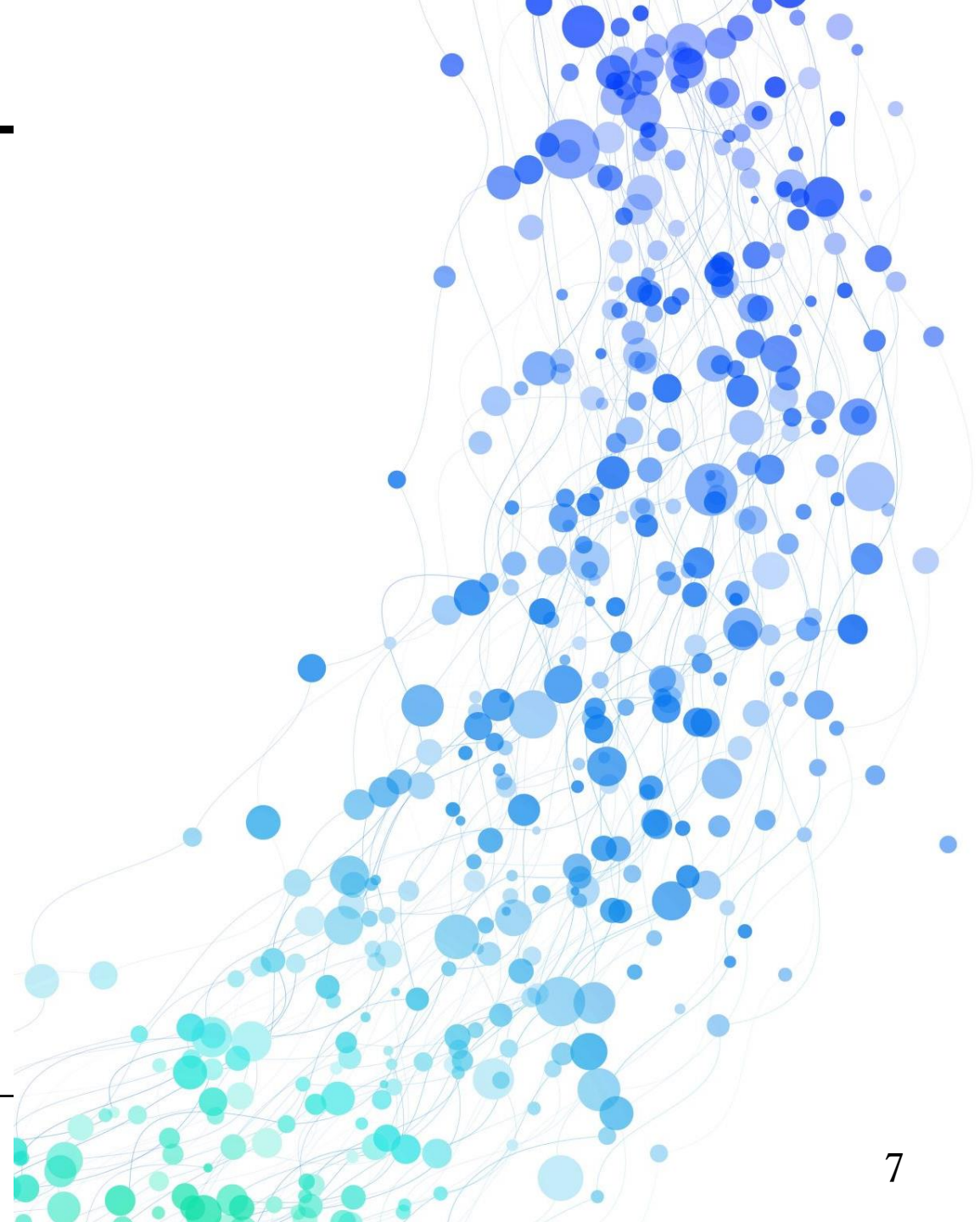
# COMMON TYPES OF MEDICAL IMAGES

- X-Rays
- Computer Tomography (CT or CAT)
- MRI (or NMR)
- PET / SPECT
- Ultrasound
- Microscopy (Whole Slide Pathology Images)

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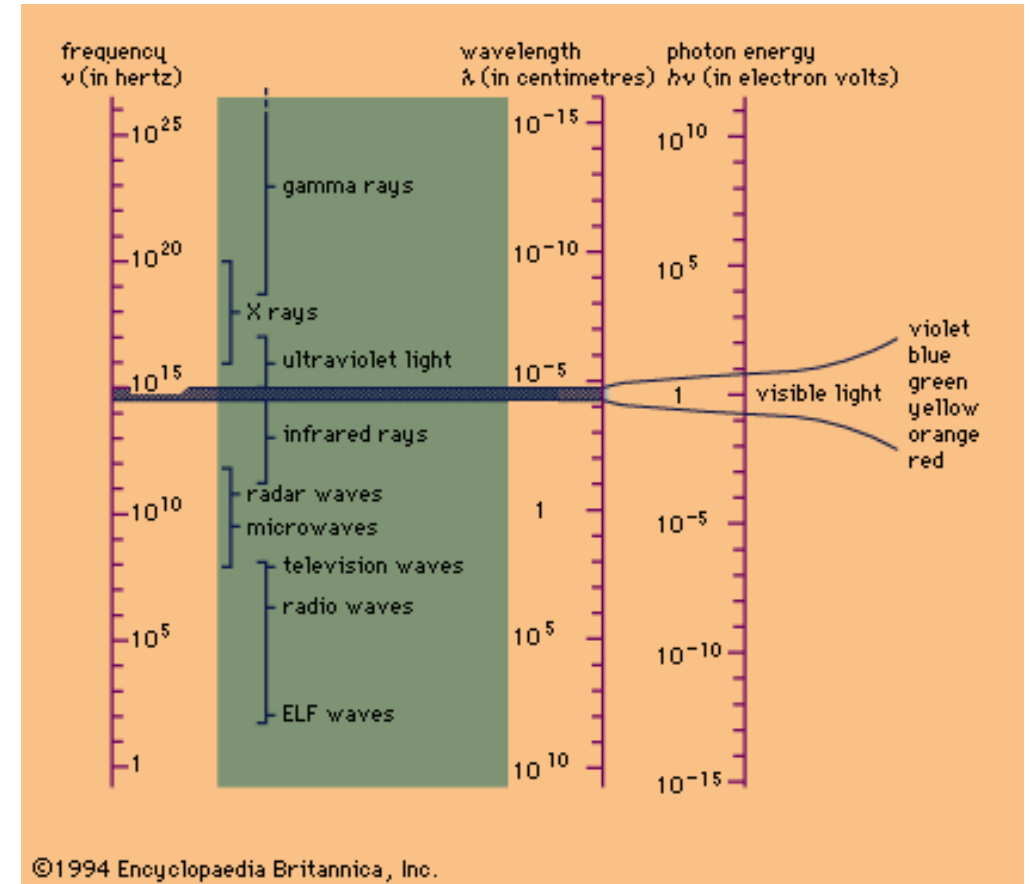
# X-RAYS

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# X-RAYS

- photons produced by an electron beam
- similar to visible light, but higher energy!





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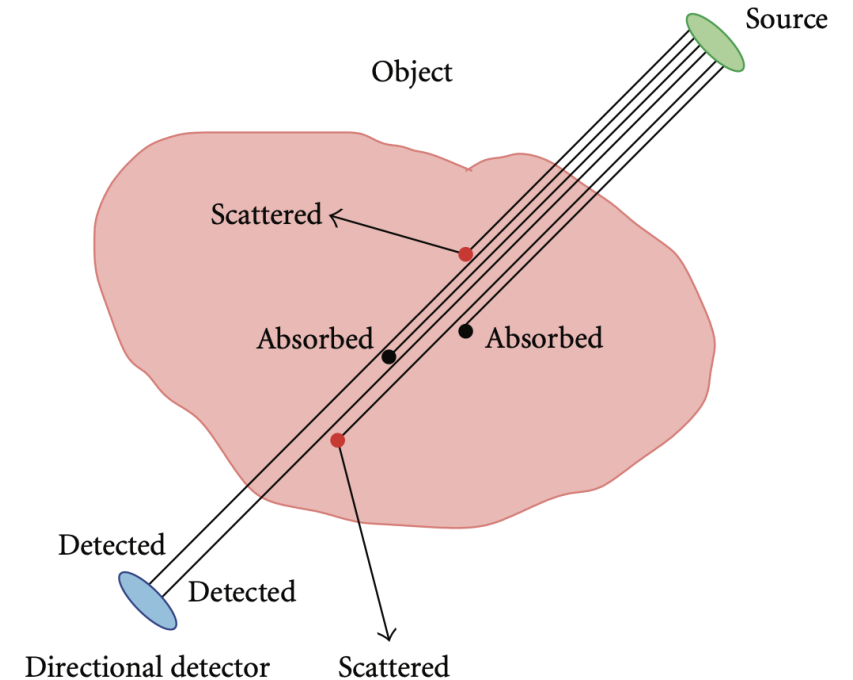
# X-RAYS - PHYSICS

- Associated with inner shell electrons
  - As the electrons decelerate in the target through interaction, they emit electromagnetic radiation in the form of x-rays.
  - patient between an x-ray source and a film → radiograph
- 👍 cheap and relatively easy to use
- 👎 potentially damaging to biological tissue

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# X-RAYS - VISIBILITY

- bones contain heavy atoms → with many electrons, which act as an absorber of x-rays
- 👉 commonly used to image gross bone structure and lungs
- 👉 excellent for detecting foreign metal objects
- 👉 main disadvantage → lack of anatomical structure
- 👉 all other tissue has very similar absorption coefficient for x-rays



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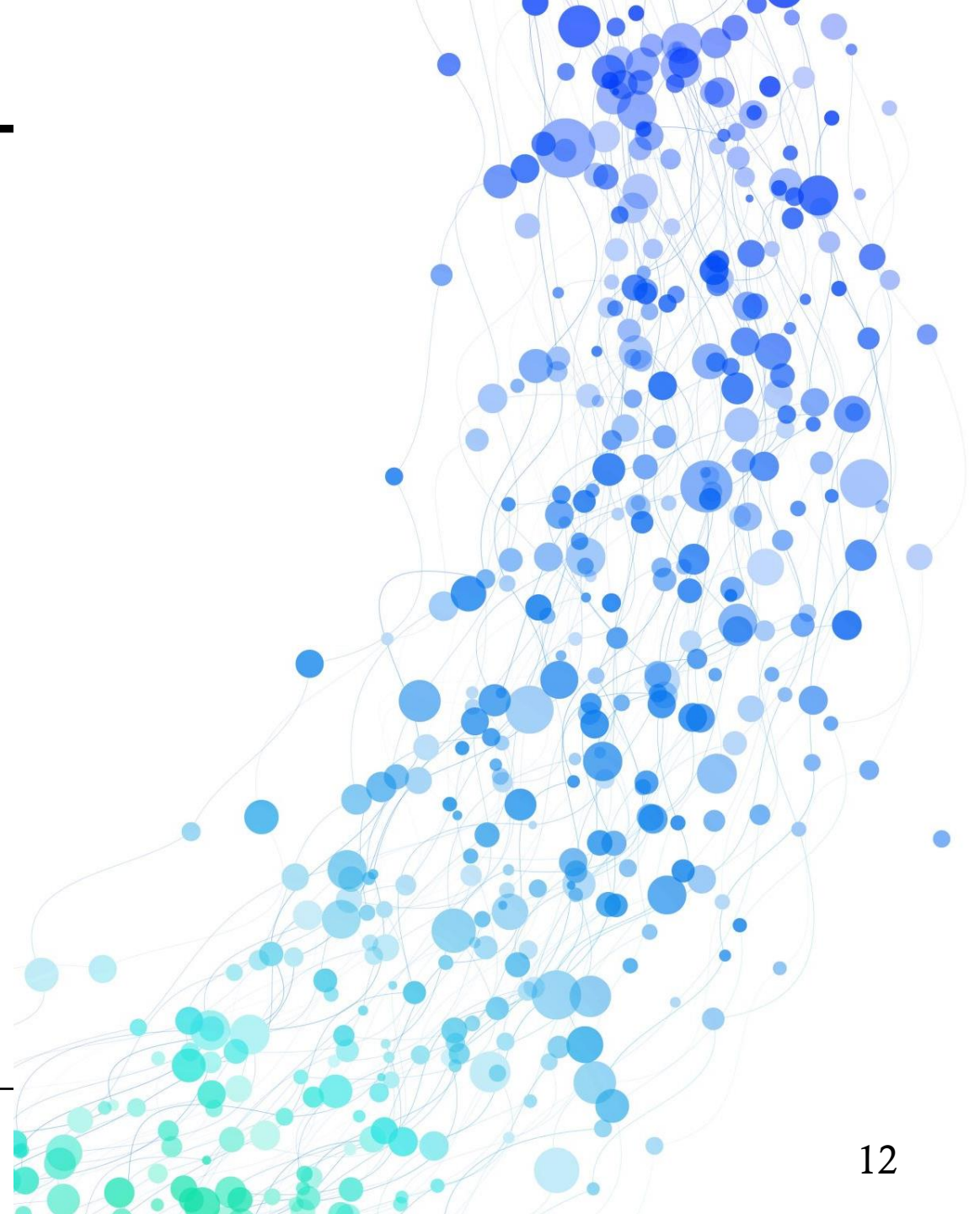
# X-RAYS - IMAGES



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# COMPUTED TOMOGRAPHY

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# CT OR CAT - PRINCIPLES

- Computerized (Axial) Tomography
- introduced in 1972 by Hounsfield and Cormack
- natural progression from X-rays
- based on the principle that a three-dimensional object can be reconstructed from its two dimensional projections
- based on the Radon transform (a map from an  $n$ -dimensional space to an  $(n-1)$ -dimensional space)





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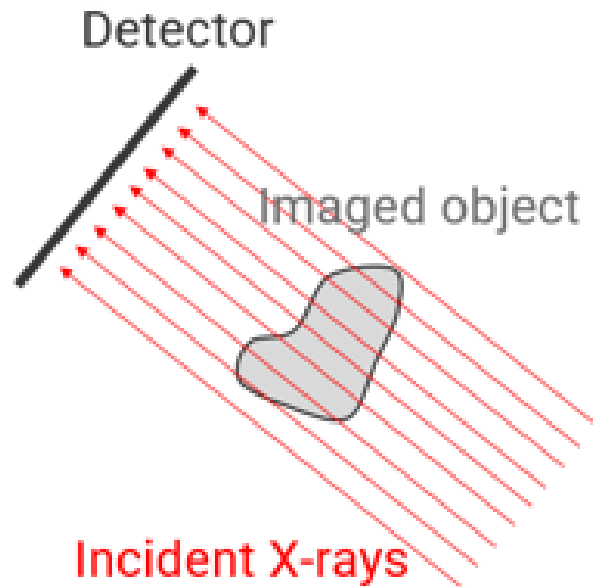
# CT OR CAT - METHODS

- measures the attenuation of X-rays from many different angles
- a computer reconstructs the organ under study in a series of cross sections or planes
- combine X-ray pictures from various angles to reconstruct 3D structures

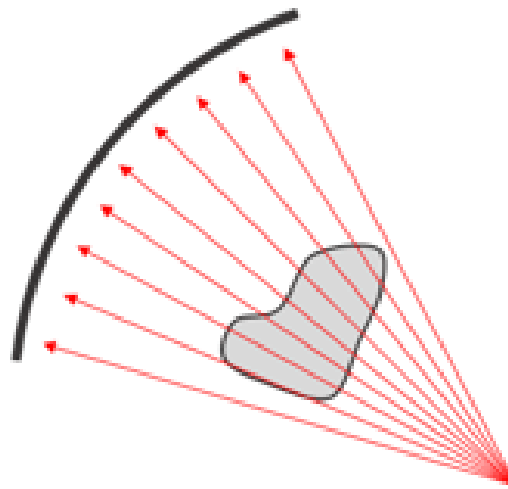


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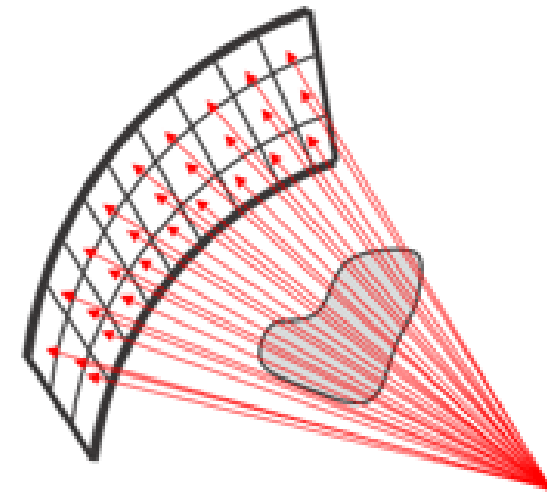
# CT - RECONSTRUCTION



Parallel beam



Fan beam

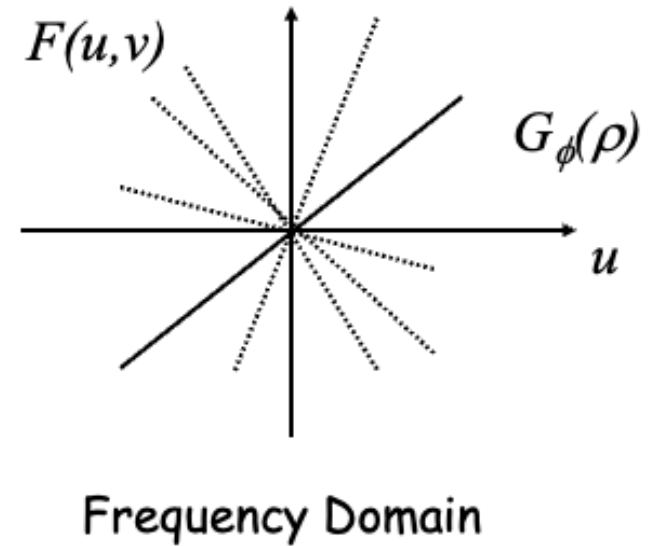
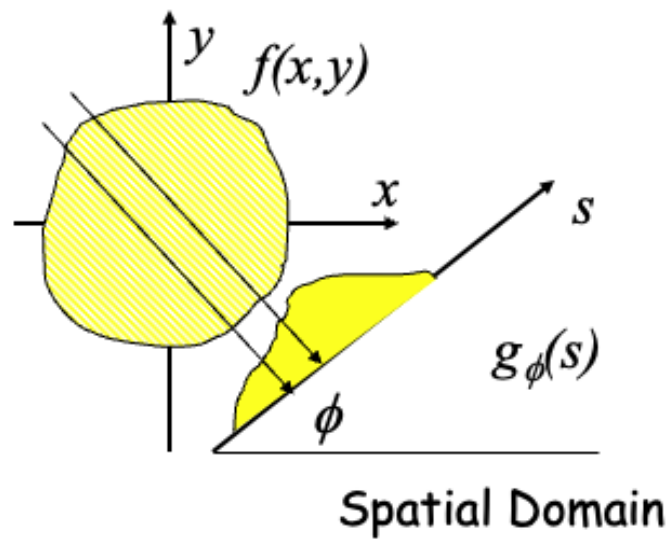


Cone beam

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# CT - RECONSTRUCTION: FBP

- **Filtered Back Projection**
- common method
- uses Radon transform and Fourier Slice Theorem

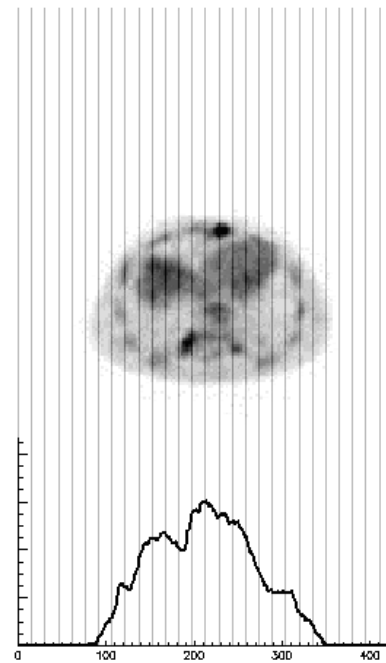


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# CT - RECONSTRUCTION: FBP

(True) Emission Volume

Sinogram (stored data)



Forward  
Projection

Theta (angle)

angle  
0

Rho (offset)

intensity profile:

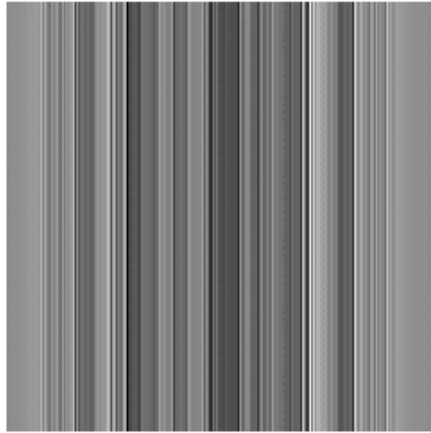


KesnersMedicalPhysics.com

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# CT - RECONSTRUCTION: FBP

Reconstructed image



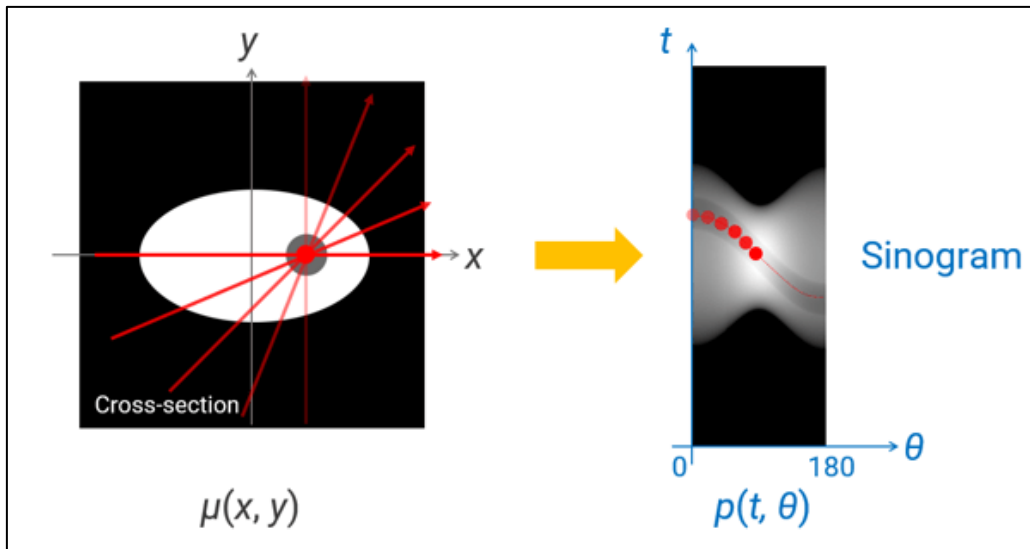
Filtered Sinogram



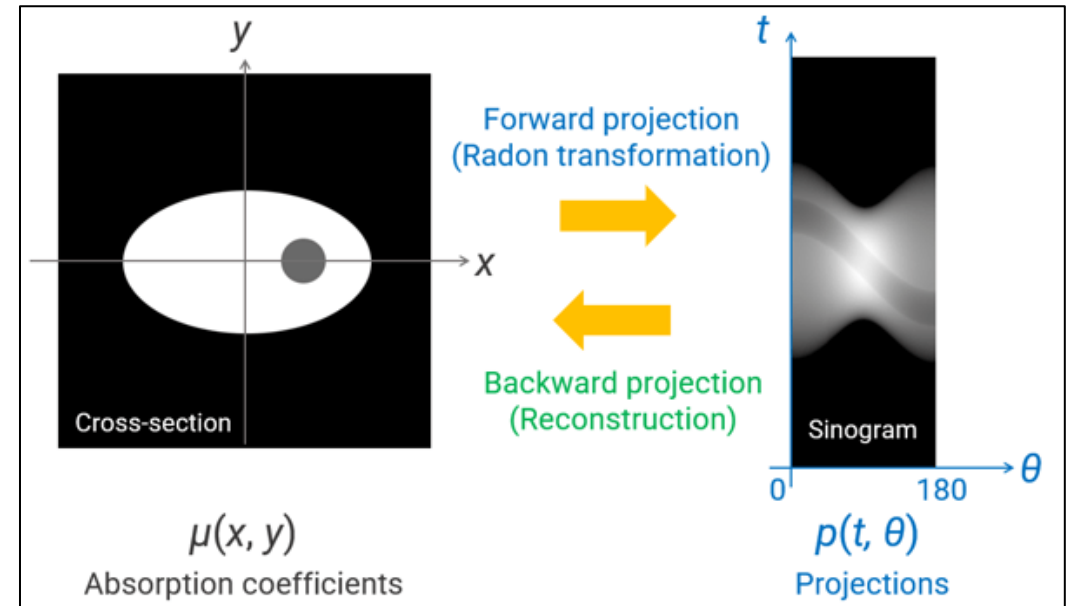
← Filtered Back Projection

KesnersMedicalPhysics.com

# CT - RECONSTRUCTION: FBP



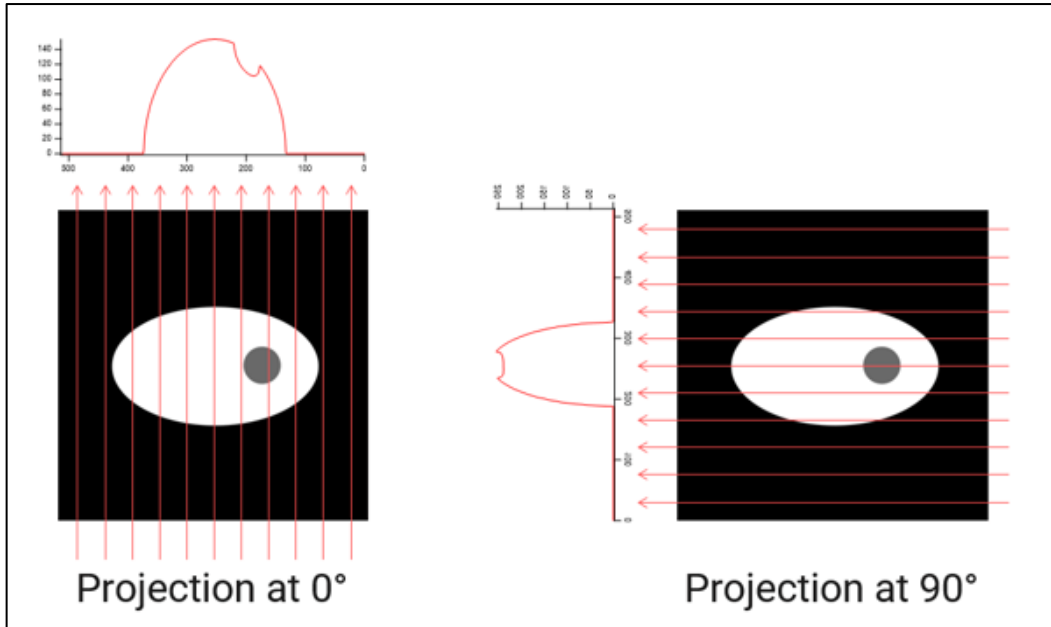
Forward Process



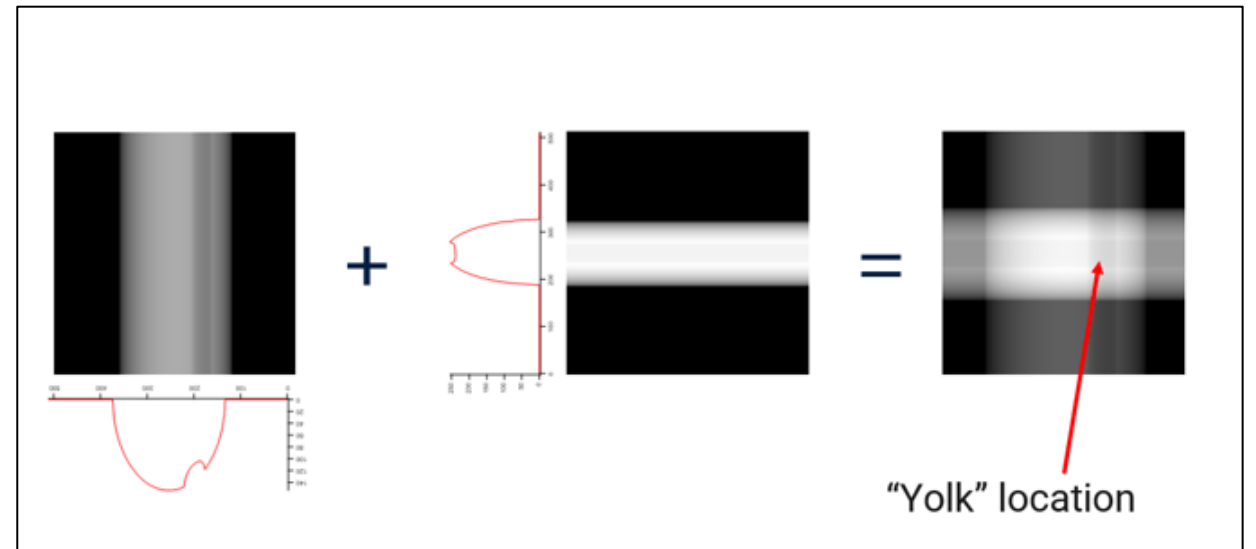
Backward Process



# CT - RECONSTRUCTION: FBP - EXAMPLE



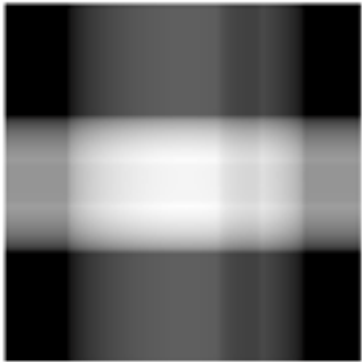
Forward Process



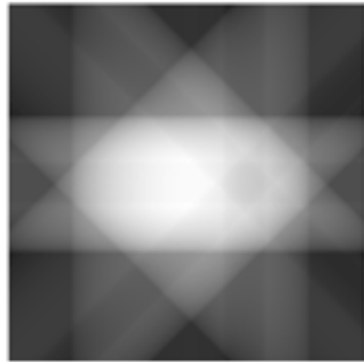
Backward Process

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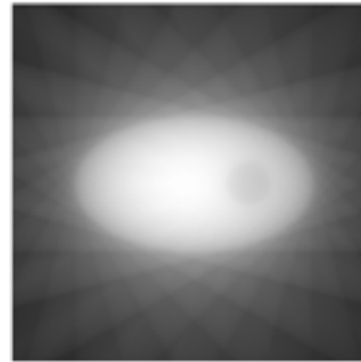
# CT - RECONSTRUCTION: FBP - EXAMPLE



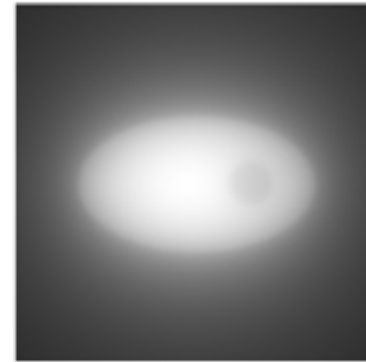
4 projections



8 projections

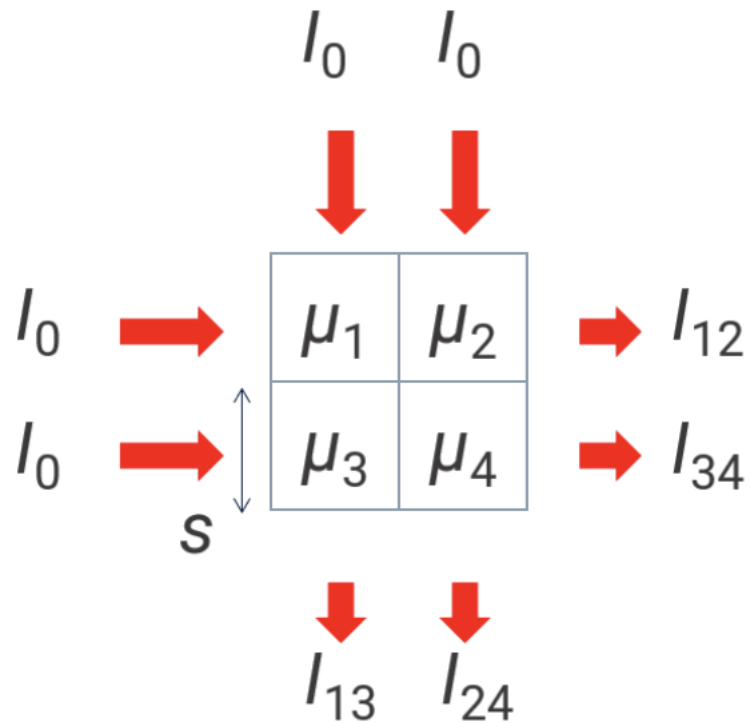


24 projections



120 projections

# CT - RECONSTRUCTION: FBP – MATH



Beer-Lambert law:

$$I = I_0 e^{-\mu s}$$

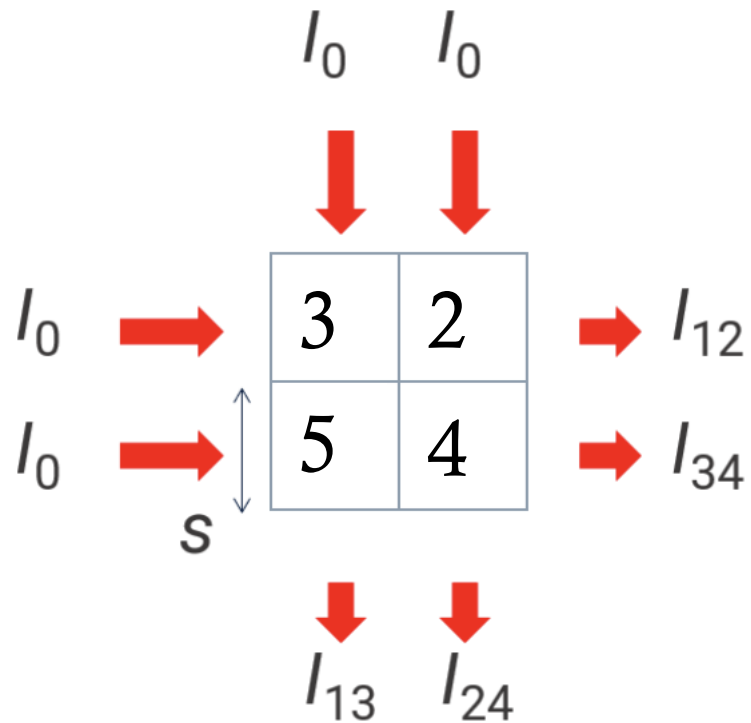
$$I_{12} = I_0 \exp\{-(\mu_1 + \mu_2)s\}$$

$$I_{34} = I_0 \exp\{-(\mu_3 + \mu_4)s\}$$

$$I_{13} = I_0 \exp\{-(\mu_1 + \mu_3)s\}$$

$$I_{24} = I_0 \exp\{-(\mu_2 + \mu_4)s\}$$

# CT - RECONSTRUCTION: FBP – MATH



Beer-Lambert law:

$$I = I_0 e^{-\mu s}$$

Given:  $I_0 = 1000, s = 1$

$$I_{12} = ?$$

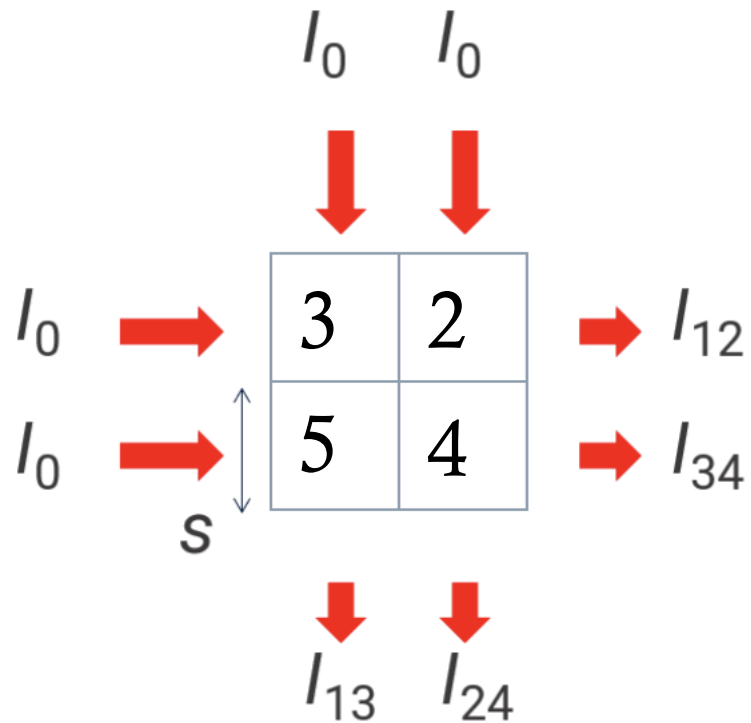
$$I_{34} = ?$$

$$I_{13} = ?$$

$$I_{24} = ?$$

And how we calculate  $\mu_1, \mu_2, \mu_3, \mu_4$ ?

# CT - RECONSTRUCTION: FBP – MATH



Beer-Lambert law:

$$I = I_0 e^{-\mu s}$$

Given:  $I_0 = 1000, s = 1$

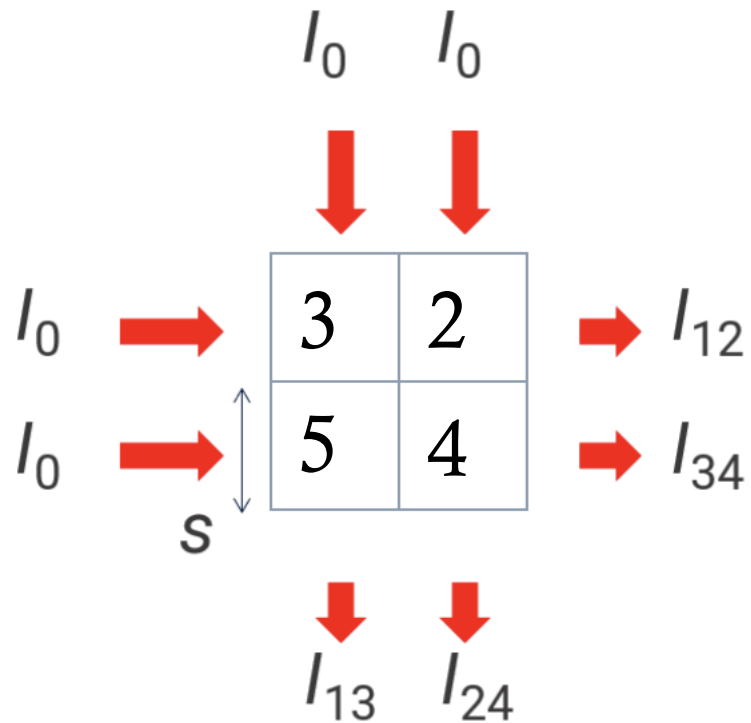
$$I_{12} = 6.738$$

$$I_{34} = 0.1234$$

$$I_{13} = 0.3355$$

$$I_{24} = 2.4788$$

# CT - RECONSTRUCTION: FBP – MATH



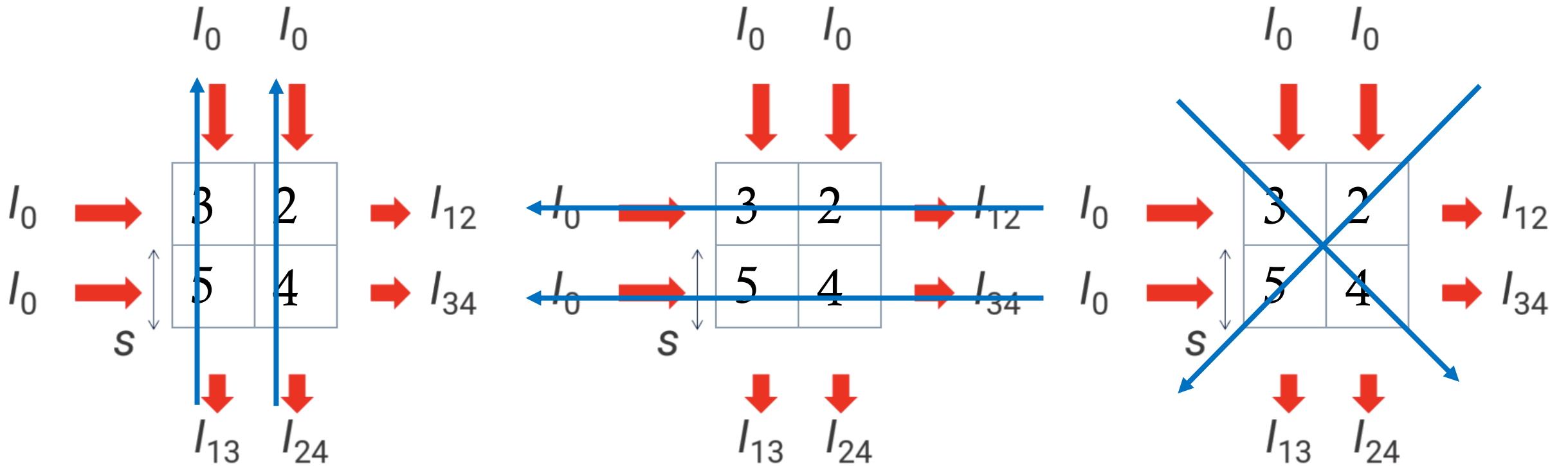
Beer-Lambert law:

$$I = I_0 e^{-\mu s}$$

$$\mu s = -\ln\left(\frac{I}{I_0}\right)$$

$$\left\{ \begin{array}{l} \mu_1 + \mu_2 = 5 \\ \mu_3 + \mu_4 = 9 \\ \mu_1 + \mu_3 = 8 \\ \mu_2 + \mu_4 = 6 \end{array} \right.$$

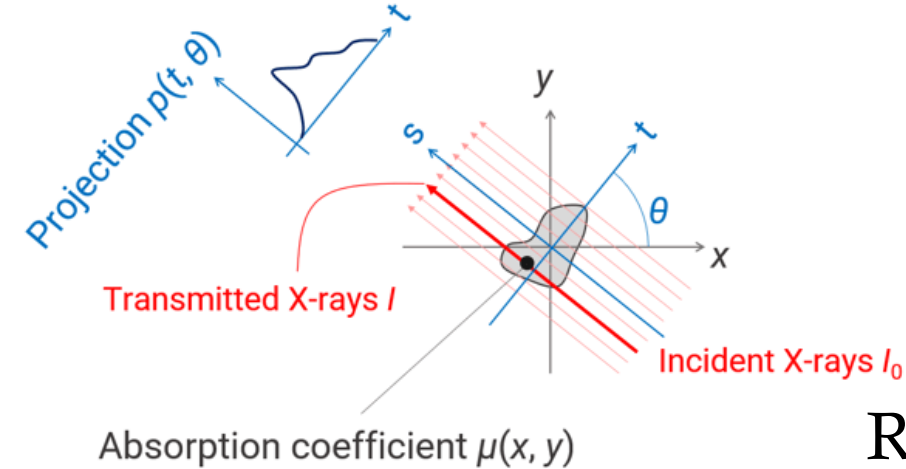
# CT - RECONSTRUCTION: FBP – EXERCISE



# CT - RECONSTRUCTION: FBP

Beer-Lambert law:

$$I = I_0 e^{-\mu s}$$



$$\frac{I}{I_0} = \exp \left[ - \int_{-\infty}^{\infty} \mu(t \cos \theta - s \sin \theta, t \sin \theta + s \cos \theta) ds \right]$$

Radon Transform:

$$p(t, \theta) = -\log \frac{I(t, \theta)}{I_0} = \int_{-\infty}^{\infty} \mu(t \cos \theta - s \sin \theta, t \sin \theta + s \cos \theta) ds$$



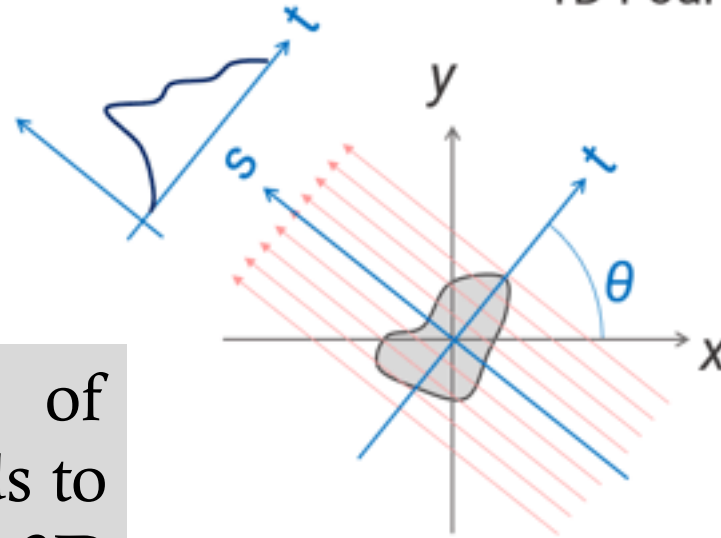
# CT - RECONSTRUCTION: FBP

$$P(\omega, \theta) = \int p(t, \theta) e^{-i2\pi\omega t} dt$$

Projection  $p(t, \theta)$

1D Fourier transform

A slice of  $F(u, v)$  at  $\theta$



Absorption coefficient  $\mu(x, y)$   
(Original image)

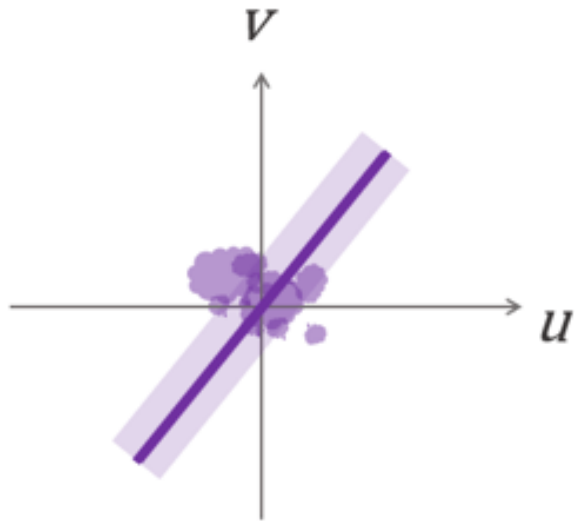
$F(u, v)$   
(2D Fourier transform of  $\mu(x, y)$ )

1D Fourier transform of projections of  $g$  corresponds to a line through the 2D or 3D Fourier transform of  $g$ .

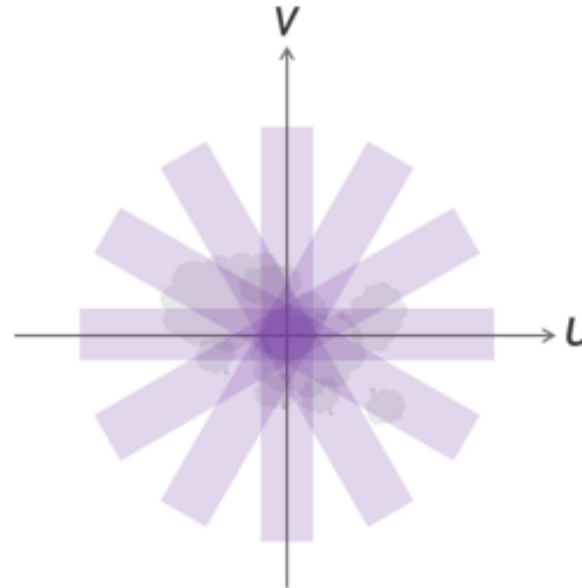
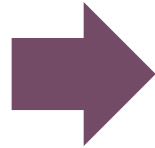
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# CT - RECONSTRUCTION: FBP

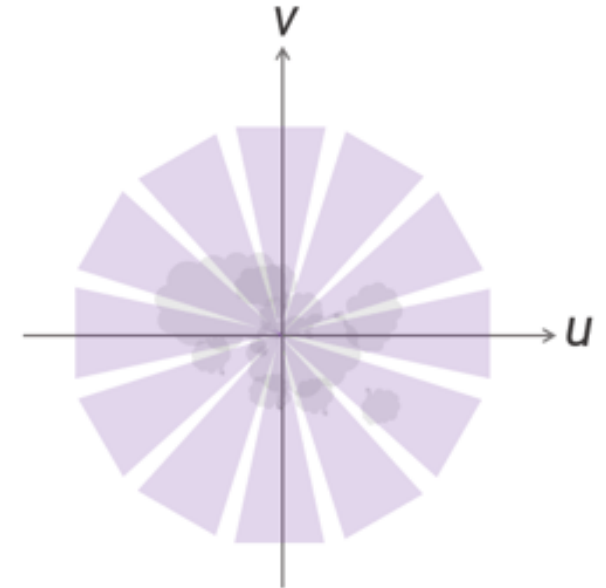
A slice of  $F(u, v)$  at  $\theta$



$F(u, v)$   
(2D Fourier transform of  $\mu(x, y)$ )



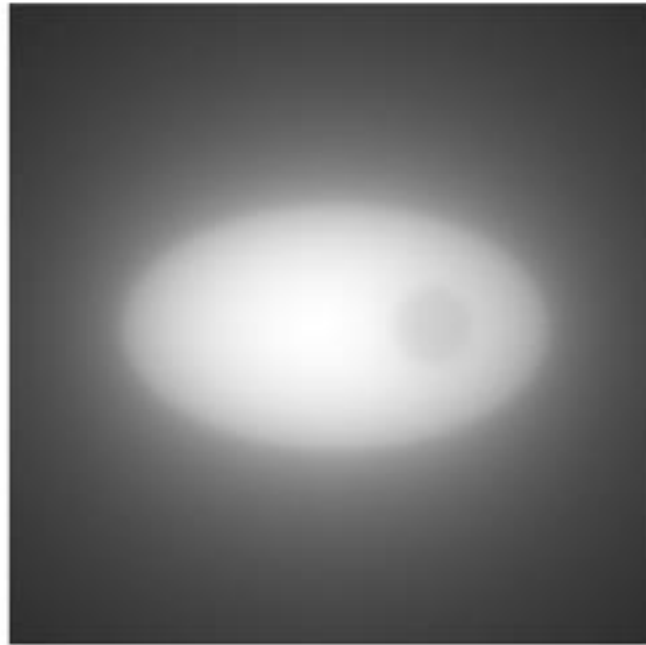
Rectangular "slices" cause  
uneven adding of slices



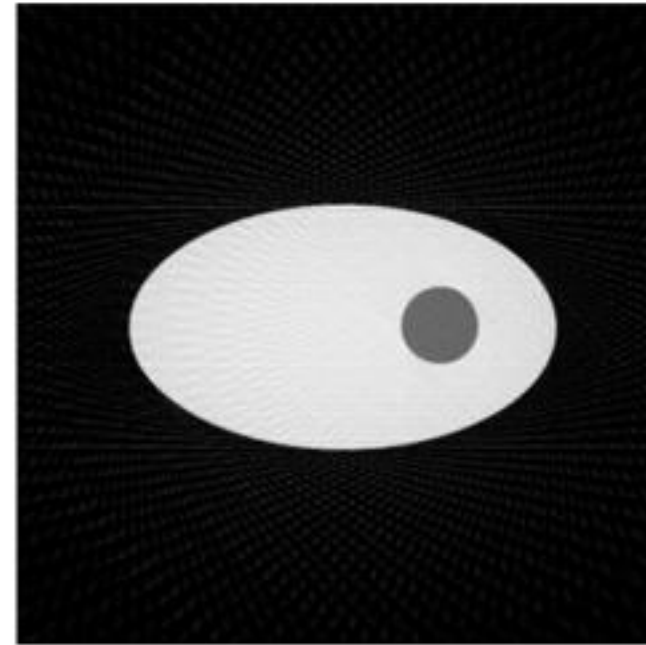
The slice shape needs to be  
evened out by "weighting"

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# CT - RECONSTRUCTION: FBP



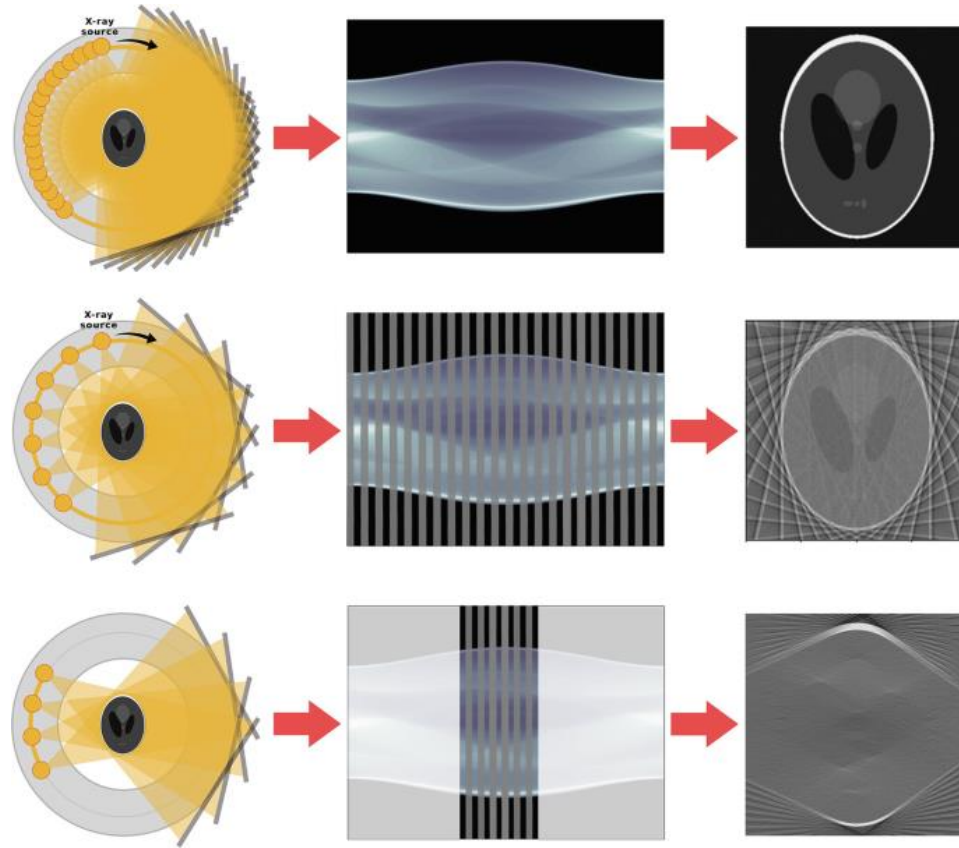
120 projections



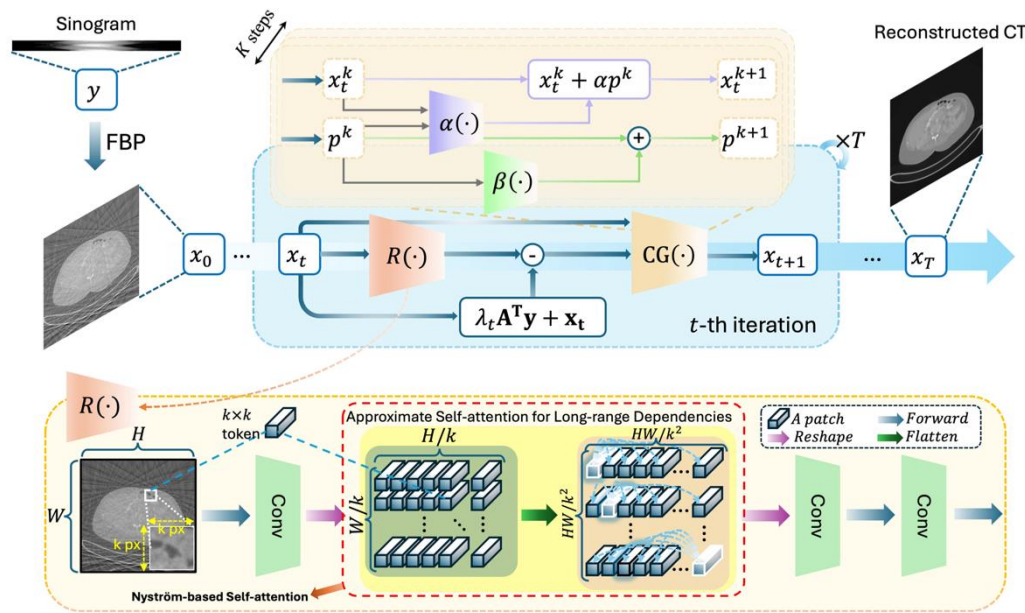
120 projections  
With ramping filter

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# CT - RECONSTRUCTION: SPARSE-VIEW



# CT - RECONSTRUCTION: SPARSE-VIEW

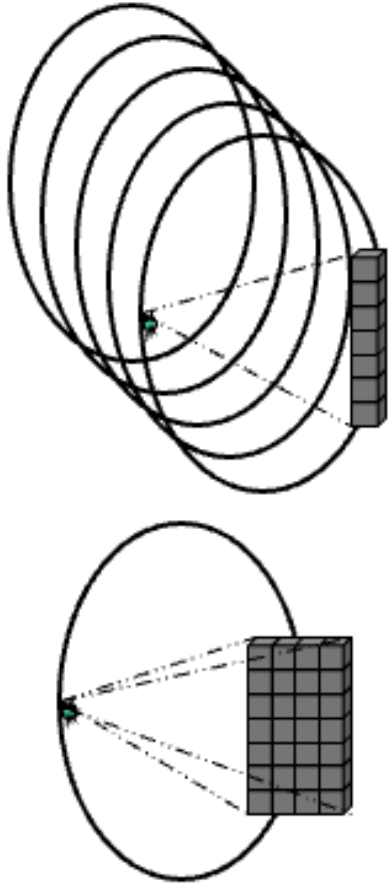


**Doanh C. Bui**, Duc-Thanh Le, Khang Nguyen, "Long-Sequence Modeling for Sparse-View Computed Tomography Reconstruction", *Machine Vision and Applications*

Tran Xuan Hieu Le, **Doanh C. Bui**, Vu Trung Duong Le, Pham Hoai Luan, Khang Nguyen, Mai K. Nguyen, Tu-Bao Ho, Yasuhiko Nakashima, "CG-GLORE: A Conjugate Gradient-Based Global-Local Regularization Network for Sparse-View CT Reconstruction", *British Machine Vision Conference*, 2026

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# CT - 2D VS. 3D



- Linear advancement (slice by slice)
  - typical method
  - tumor might fall between 'cracks'
  - takes long time
- helical movement
  - 5-8 times faster
  - A whole set of trade-offs

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# CT OR CAT - ADVANTAGES

- 👍 significantly more data is collected
  - 👍 superior to single X-ray scans
  - 👍 far easier to separate soft tissues other than bone from one another (e.g. liver, kidney)
  - 👍 data exist in digital form → can be analyzed quantitatively
  - 👍 adds enormously to the diagnostic information
  - 👍 used in many large hospitals and medical centers throughout the world
-

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# CT OR CAT - DISADVANTAGES

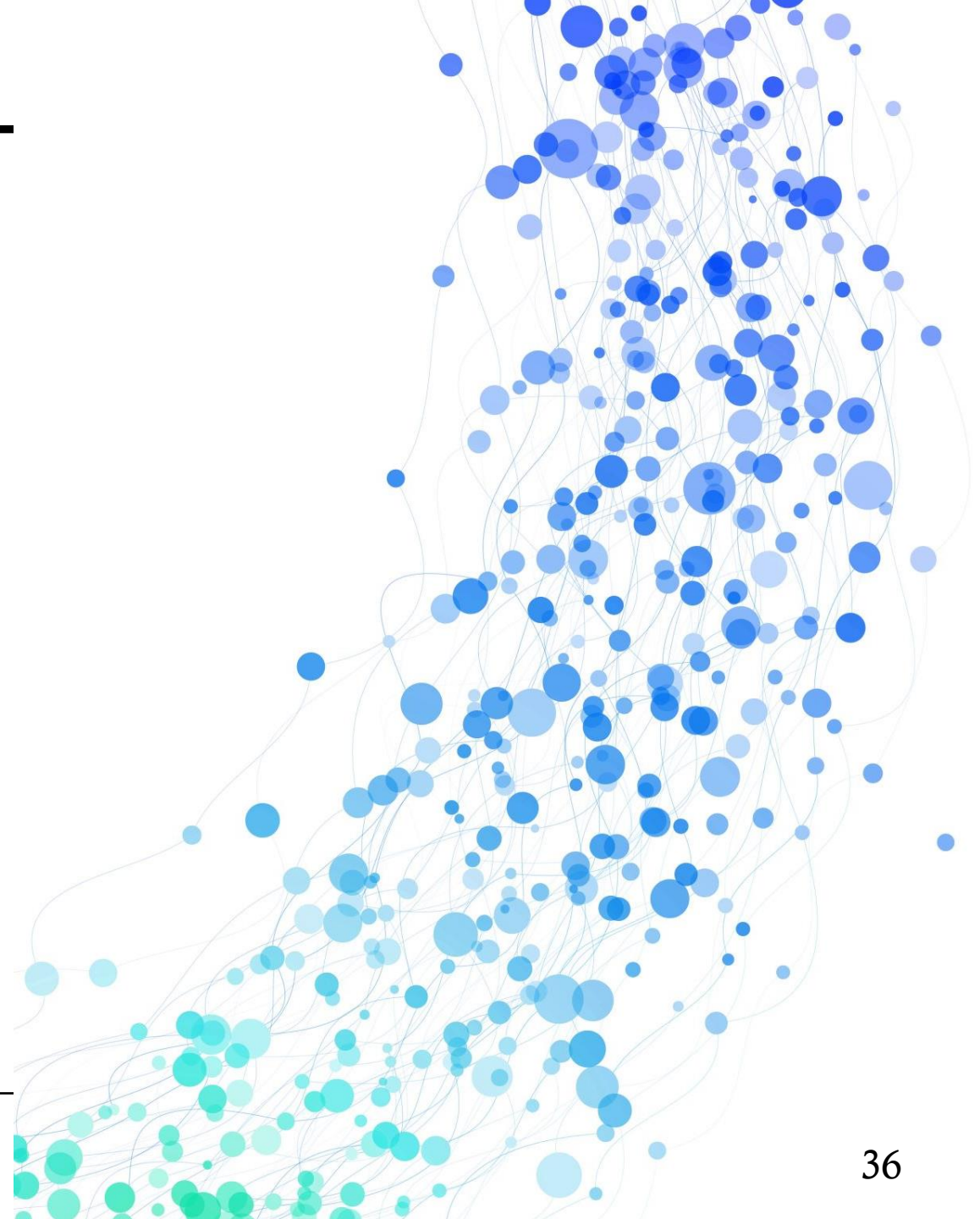
- 👎 significantly more data is collected
  - 👎 soft tissue X-ray absorption still relatively similar
  - 👎 still a health risk
- MRI is used for a detailed imaging of anatomy



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# MAGNETIC RESONANCE IMAGING

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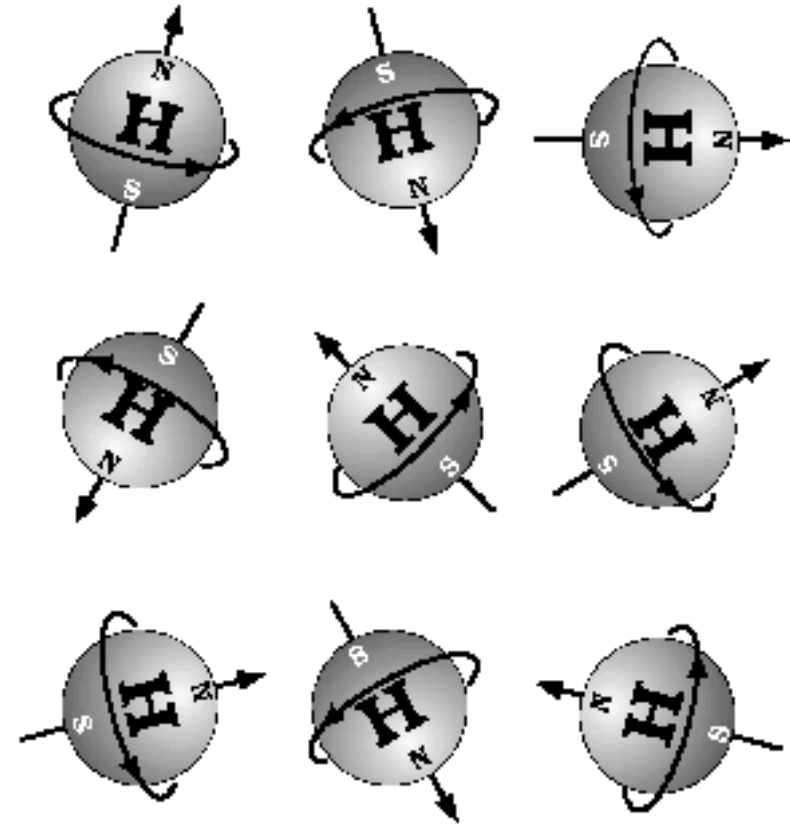
# MRI

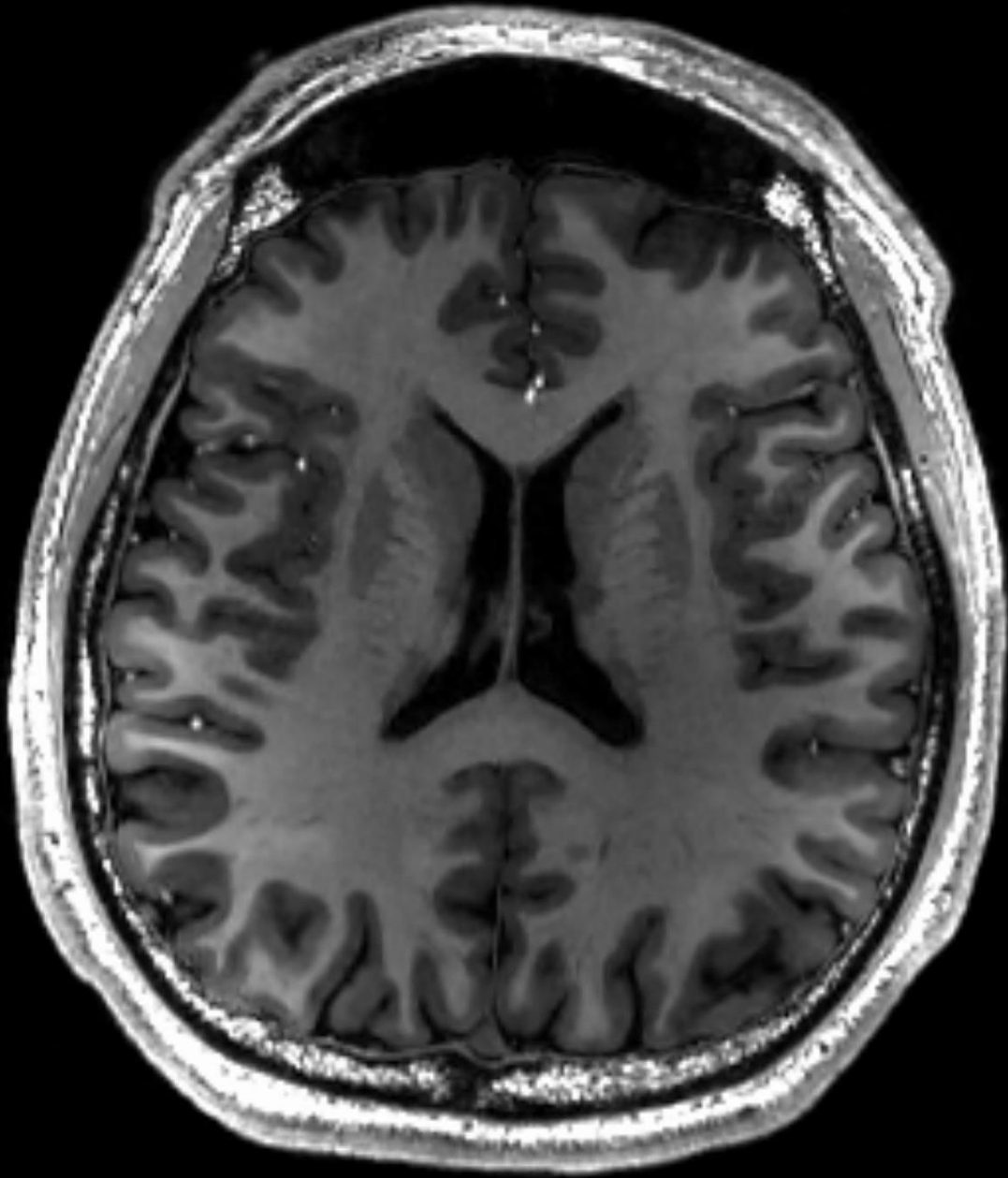
- Nuclear Magnetic Resonance (NMR) (or Magnetic Resonance Imaging - MRI)
  - most detailed anatomical information
  - high-energy radiation *is not* used, i.e. “safe”
  - based on the principle of nuclear resonance
  - (medicine) uses resonance properties of protons
-

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# MRI - POLARIZED

- all atoms (core) with an odd number of protons have a 'spin', which leads to a magnetic behavior
- Hydrogen (H) - very common in human body + very well magnetizing
- Stimulate to form a macroscopically measurable magnetic field





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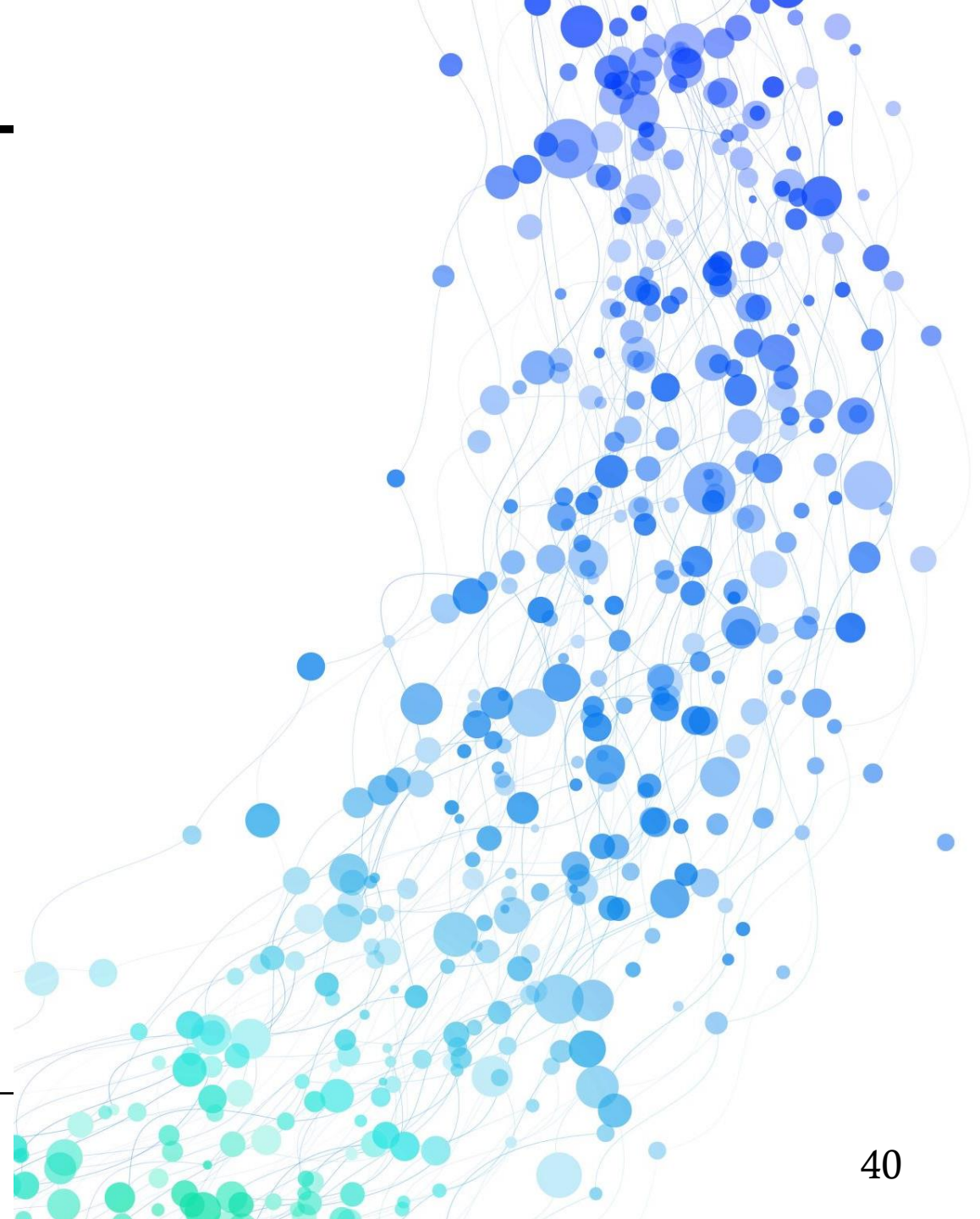
# MRI - SIGNAL TO NOISE RATIO

- proton density pictures - measures H  
MRI is good for tissues, but not for bone
- signal recorded in Frequency domain!!
- Noise - the more protons per volume unit, the more accurate the measurements - better SNR through decreased resolution

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# PET/SPECT

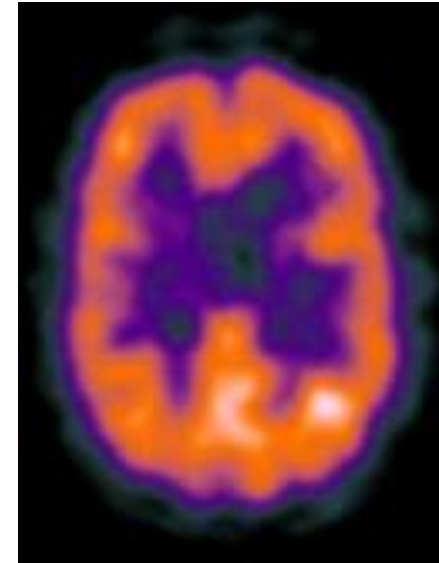
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# PET/SPECT

- Positron Emission Tomography  
Single Photon Emission Computerized Tomography
- recent technique
- involves the emission of particles of antimatter by compounds injected into the body being scanned
- follow the movements of the injected compound and its metabolism
- reconstruction techniques similar to CT - Filter Back Projection & iterative schemes

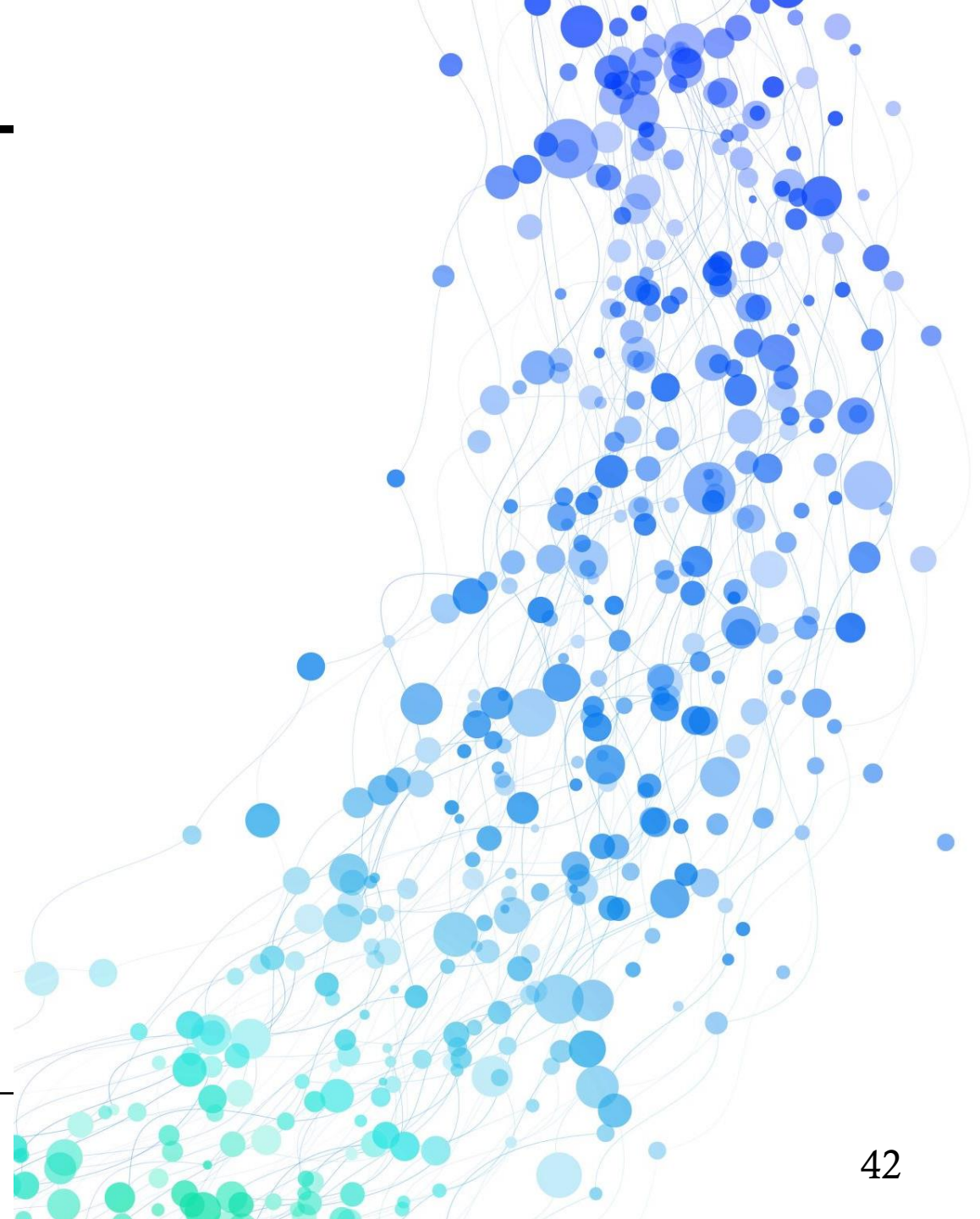




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# ULTRASOUND

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# ULTRASOUND

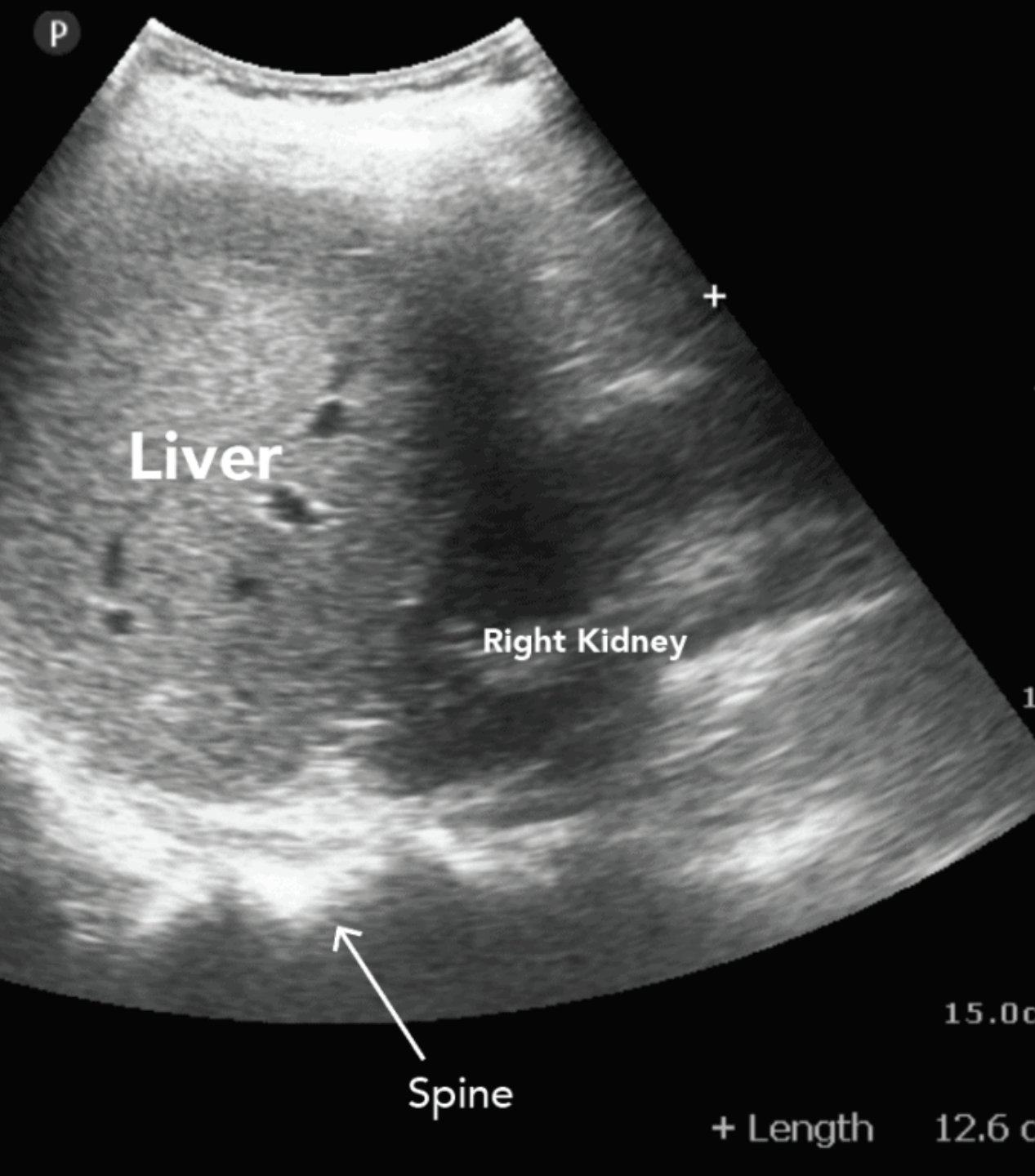
- the use of high-frequency sound (ultrasonic) waves to produce images of structures within the human body
  - above the range of sound audible to humans (typically above 1MHz)
  - piezoelectric crystal creates sound waves
  - aimed at a specific area of the body
  - change in tissue density reflects waves
  - echoes are recorded
-



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# ULTRASOUND (2)

- Delay of reflected signal and amplitude determines the position of the tissue
  - still images or a moving picture of the inside of the body
  - there are no known examples of tissue damage from conventional ultrasound imaging
  - commonly used to examine fetuses in utero in order to ascertain size, position, or abnormalities
  - also for heart, liver, kidneys, gallbladder, breast, eye, and major blood vessels
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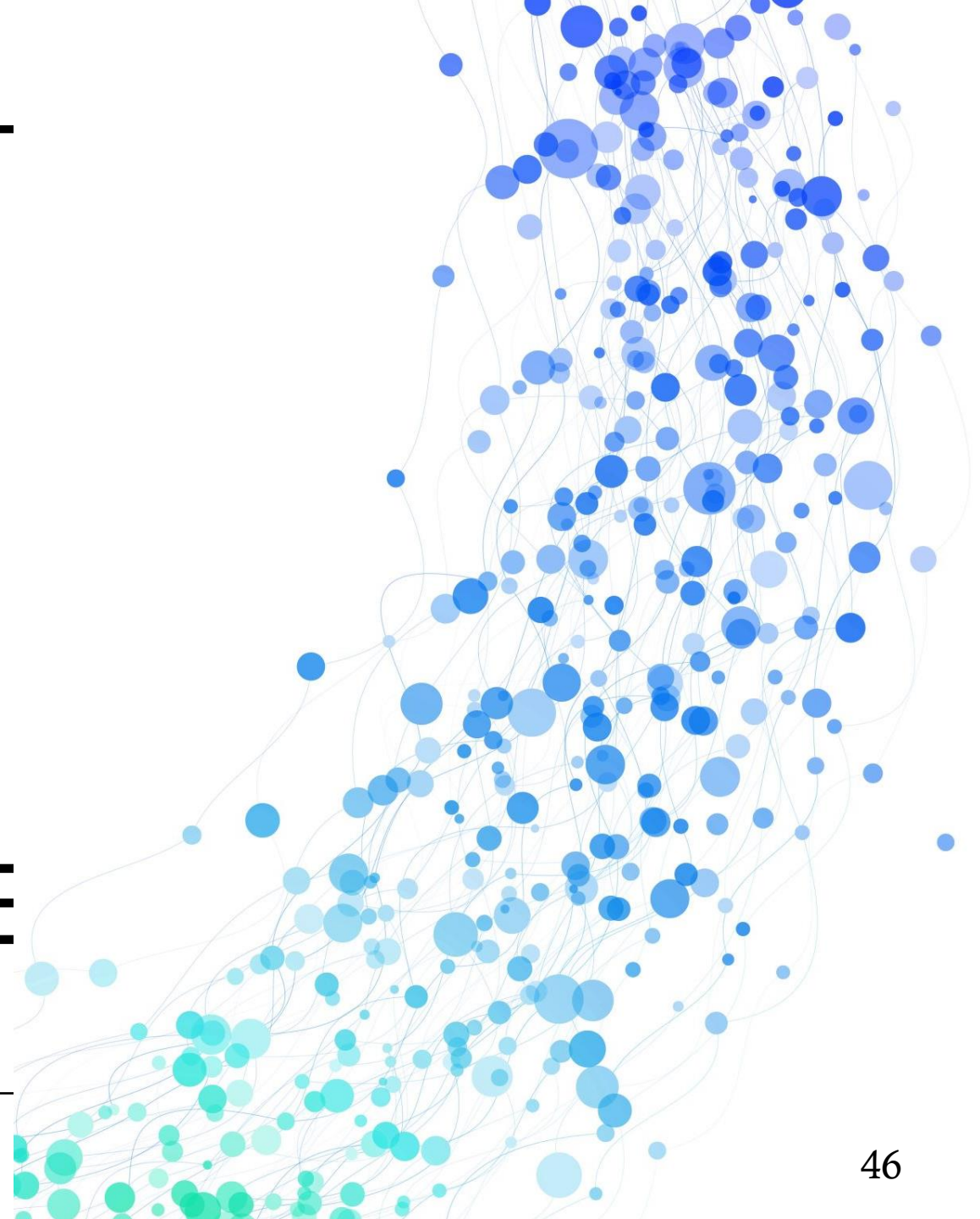
## ULTRASOUND (3)

- by far least expensive
  - very safe
  - very noisy
  - 1D, 2D, 3D scanners
  - irregular sampling - reconstruction problems
-

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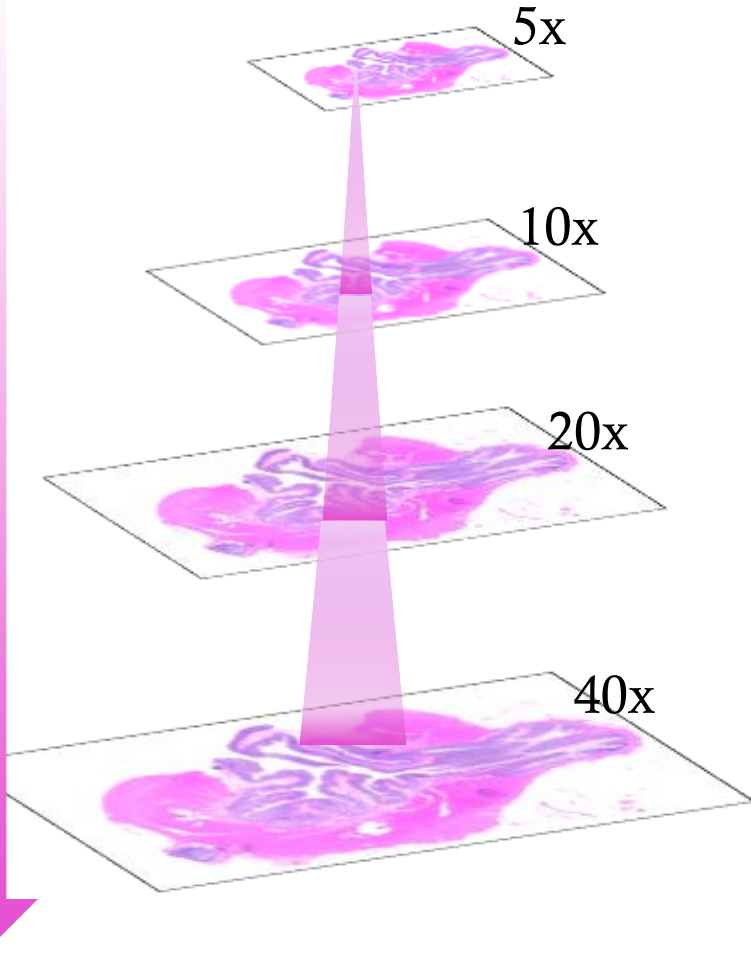
# MICROSCOPY (WHOLE SLIDE IMAGES)

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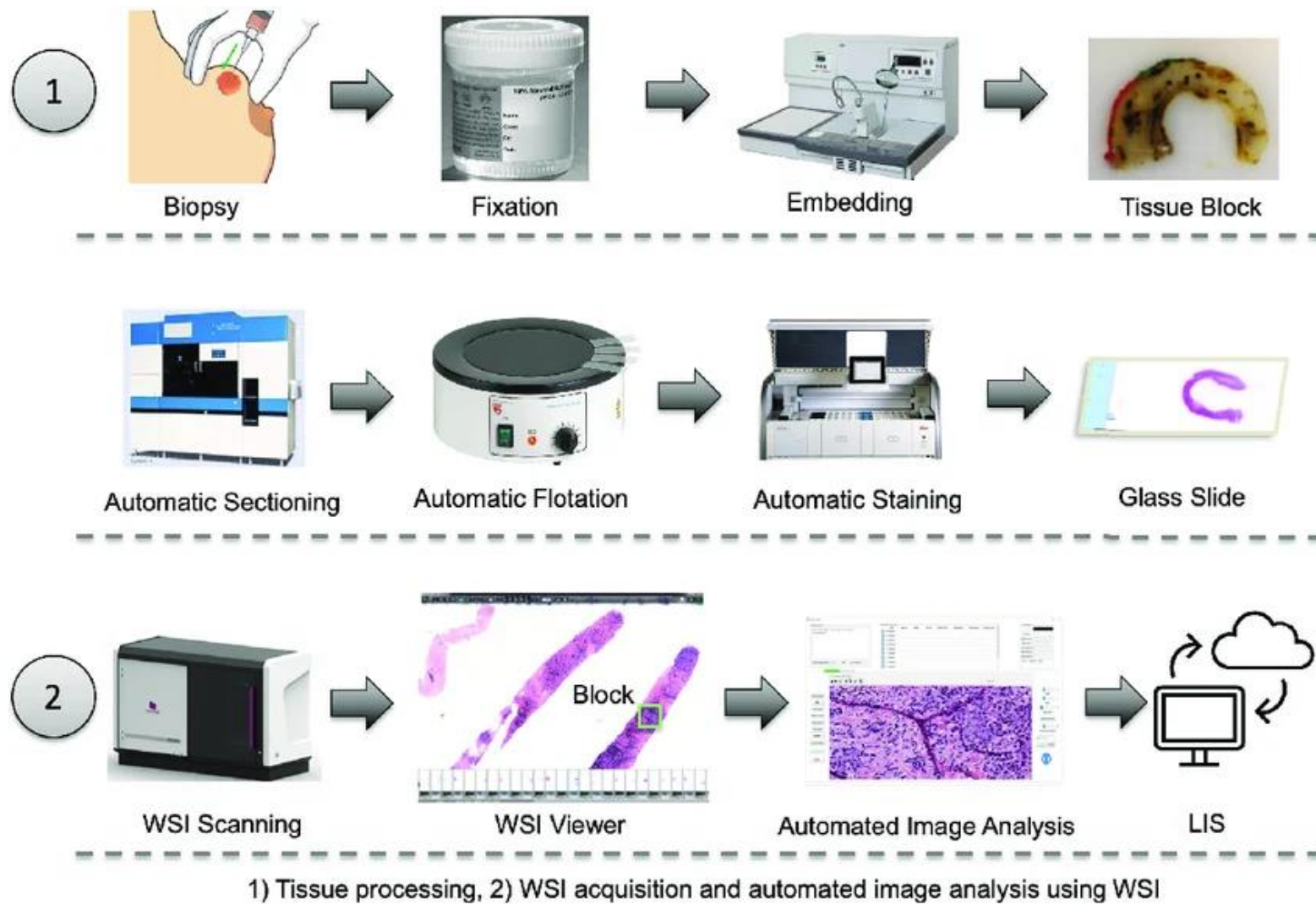


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# PATHOLOGY IMAGING



- Pathology Imaging refers to the process of tissue visualization and analysis, often carried out through invasive means like biopsies.
- Typically, WSIs are stored in a pyramid structure containing multiple magnifications, ranging from low to high.
- Digital computational methods, particularly those leveraging advanced AI techniques, have been developed to process WSIs for specific tasks, such as cancer classification and sub-typing, tissue/nuclei/gland segmentation, and survival analysis.





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*Thank  
you!*

**THANK YOU!**

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